

Functional Food Potential of Flaxseed Oil-Fortified Granola Bars: Nutritional Enhancement and Sensory Evaluation"

. Flaxseed oil essential nutrients

Dietary fibers

Flax fibers are one of the oldest fiber crops in the world. Linen, which is made from flax, dates back at least to ancient Egyptian times. The skin of the plant's stem is where flax fiber comes from. The total flax plant consists of about 25 % seed and 75 % stem and leaves (Mohit Chaudhary & Poonia). The stem or non-seed parts are approximately 20% fiber, which can be extracted through chemical or mechanical retting. The flax fiber is a natural and biodegradable composite, which has good mechanical properties and low density (Raghuwanshi, Agrawal, & Mane, 2019). Flax fiber is lustrous, flexible and soft, ; strands of the stuff resemble blonde hair, hence the term “flaxen.” Stronger than cotton fiber but less elastic (Pavelek, Tejklová, & Bjelková, 2015). Flax fiber is also a source material for the manufacture of fine paper for printing bank notes, and of rolling paper for cigarettes and tea bags. Flax fibers are also known as soluble and insoluble dietary fibers. The ratio of soluble fiber to insoluble fiber is between 20:80 and 40:60. Lignin and Cellulose make up the principal insoluble fiber fraction, and the mucilage gums are the soluble fiber fractions (Kürkçü & Dedeoğlu, 2022; Soni, Katoch, Kumar, & Verma, 2016).

Water extraction produces mucilage and this substance functions well as a foam stabilizer. The laxative properties of mucilage gums are dependent on their ability to become viscous when combining with other fluids or water because they are soluble polysaccharides (Soni et al., 2016). The viscosity of flaxseed mucilage reaches optimal levels within the pH environment of human intestines which ranges from 6–8. Consuming 10 grams of flaxseed per day provides the body with one gram of soluble fiber along with three grams of insoluble fiber (Hinduja et al., 2023). The mechanism behind insoluble fiber's laxative benefits includes enhanced stool bulk and shortened bowel passage duration which secures effective laxation and prevents bowel strain. (Gutte, Sahoo, & Ranveer, 2015).

Alpha-linolenic acid (ALA),

A plant-based omega-3 fatty acid is enriched with ALA. Flax seeds are a common source of alpha-linolenic acid from flax seeds; about 55% of oil is extracted. Although it is an essential fatty acid,

the human body obtains it from the diet; it cannot be synthesized in the body(Ogawa, Sawane, Ookoshi, & Kawashima, 2023). After the ingestion, it is converted into DHA (docosahexaenoic acid) and EPA (eicosapentaenoic acid), long-chain fatty acids. Despite that ALA (1%) is transformed into a physiological amount of DHA and EPA, revealing that this conversion process is ineffective. ALA is associated with different health benefits, such as reducing inflammation, which is helpful in rheumatoid arthritis(Hussain, Kaur, & Mohapatra, 2021; Isham, Hewison, & Bradbury-Jones, 2019; Ogawa et al., 2023).

In addition, some research studies have found that ALA also reduces the risk of cardiovascular disease. One tablespoon of flaxseed oil provides a significant amount of omega-3 fatty acids(Yang et al., 2021).

Preparation of Flaxseed oil

Selection of Flaxseeds

To achieve high-quality oil extraction, flaxseeds with adequate moisture content were chosen. By maintaining controlled moisture content, avoid microbial growth and guarantee maximum oil extraction. The flaxseeds were obtained from a reliable source to ensure cold-pressing specifications(1, 2016 ; Tańska, Roszkowska, Skrajda, & Dąbrowski, 2016).

Cleaning and Removal of Contaminants

Flaxseeds were thoroughly cleaned from dirt, contaminants, and dust to remove lighter and heavier impurities, mostly using air aspiration and mechanical sieving techniques before oil processing. This step is important to prevent contaminants and impurities from affecting oil quality.

Washing and Drying

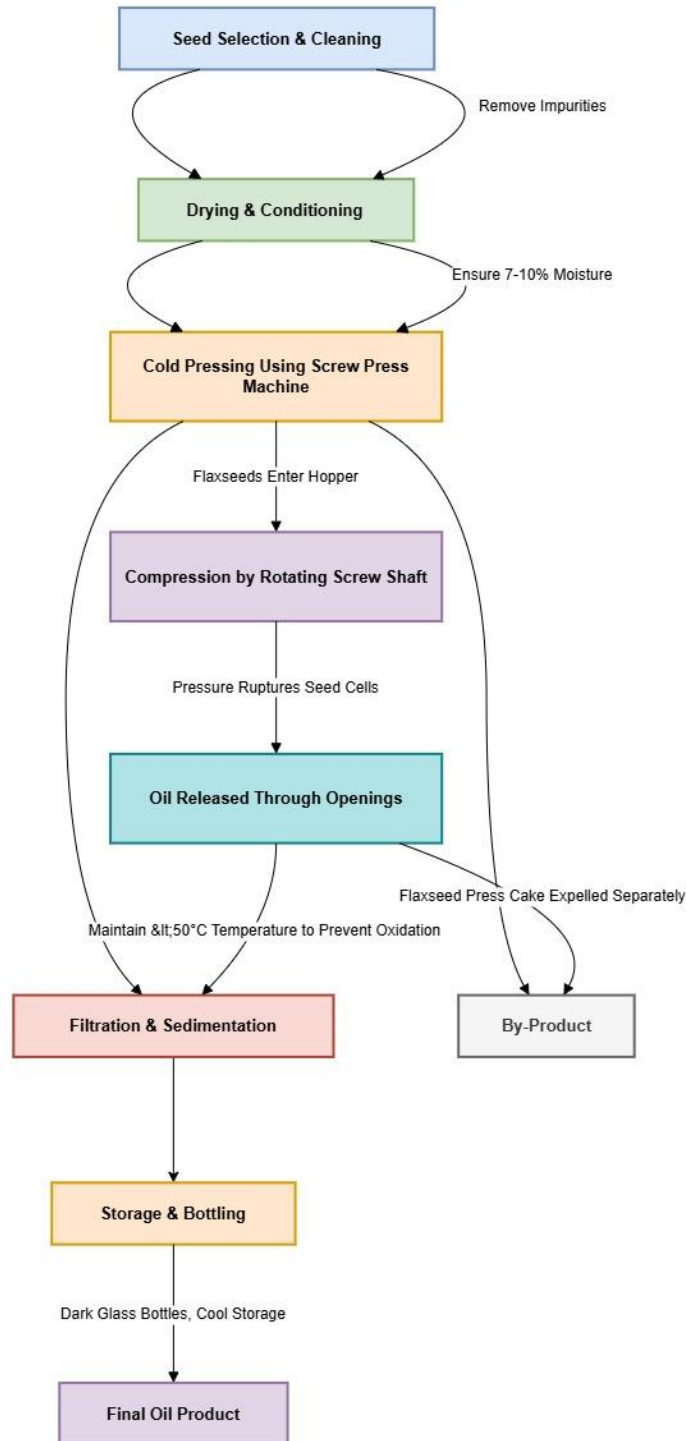
In the last step to get rid of any remaining marks, flaxseeds were rinsed with water. This process was performed in controlled environment conditions to avoid any contamination. After the cleaning process, the flaxseeds were dried to maintain moisture content (6–8%) in the drying chamber at low temperature (less than 50 °C). For effective oil extraction and to stop microbial growth, an adequate drying process is necessary.

Cold Pressing Process

A screw pressing machine that has a revolving screw shaft within a cylindrical pressing chamber used to perform the extraction process. Through the hopper, flaxseeds were added in the chamber and then gradually compressed, and screw rotation moved them forward. This high pressure

ruptures the flaxseed cells, and the flaxseed cake was drained separately while the oil was collected through a tiny hole in the chamber (Ferreira et al., 2024). It is essential to maintain a low Extraction heat because excessive temperatures above 50°C may cause degradation and oxidation of bioactive compounds.

Oil Extraction Process



Digital sensors or infrared thermometers were used to measure and monitor control temperature and ensure that frictional heat produced while the pressing process remains within an acceptable range. To avoid any large particles, the extracted oil was first filtered with filter cloth or fine mesh. To increase the purity of oil, a second filtration process of centrifugation or vacuum filtration was used (Wandhekar, Pawar, Shinde, & Gangakhedkar, 2023).

Oil Yield and Storage

The oil extraction yield was calculated with the following formula:

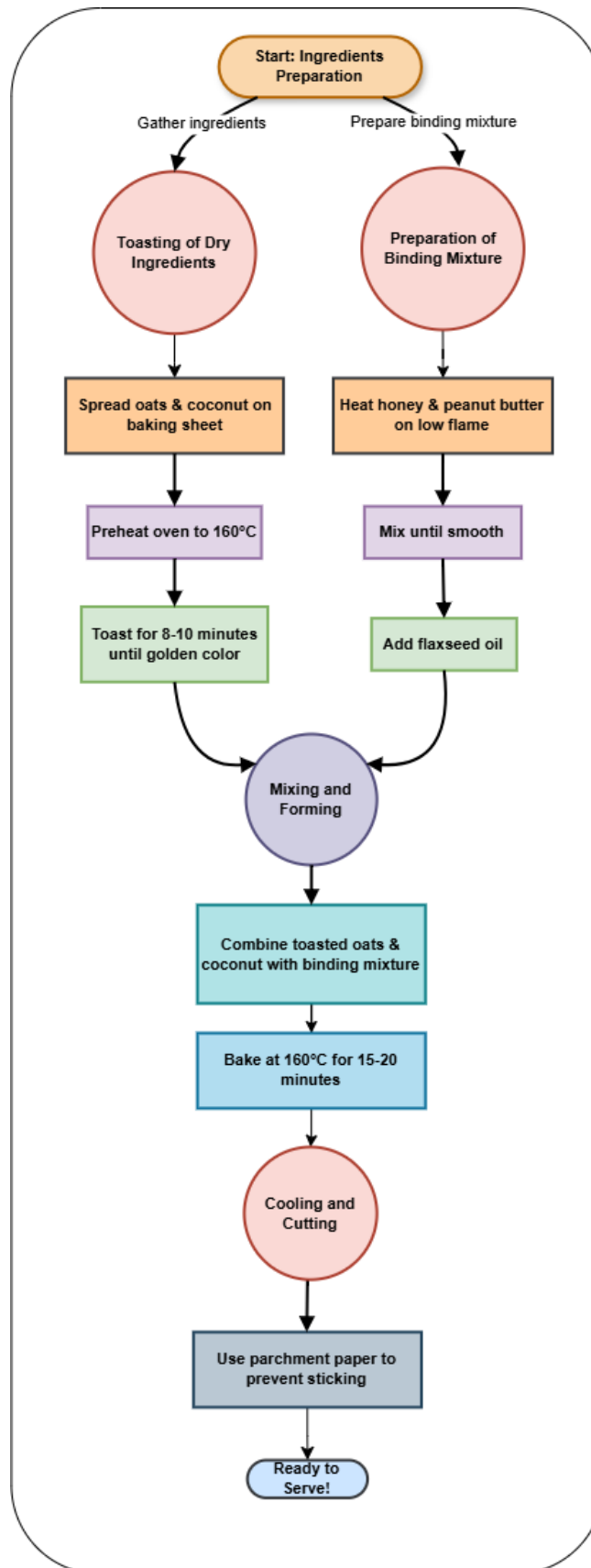
$$\text{Oil yield (\%)} = \left(\frac{\text{weight of the extracted oil}}{\text{weight of flaxseed used}} \right) \times 100$$

Storage of Extracted Oil

After the filtration and extraction process, proper storage is necessary to prevent oxidation and rancidity and maintain oil freshness; it is kept in airtight glass bottles. The oil bottle was stored in a dark and cool place to avoid oxidative degradation, as exposure to light and oxygen activates the oxidation process.

Flaxseed oil extraction byproduct

The oil extraction byproduct is also known as flaxseed press cakes, which are enriched with fiber, lignans, and protein. Instead of discarding it, it is used as a bioactive compound for medicinal products and as a functional food in animal feed (Shim, Gui, Wang, & Reaney, 2015).



Physicochemical Testing

Moisture Content

The moisture content of flaxseed oil enriched granola bars was determined as per the standard methods of AOAC 2000. A representative portion of about 5g granola bars was taken and placed in a dish or crucible that was already weighted. Then the sample with the crucible was weighed, and the W1 (initial weight) was noted. To ensure that all moisture evaporated, the sample was placed in a hot air oven (105°C) for 4 to 6 hours. The sample was dried and then allowed to cool at room temperature in a desiccator before W2 weight. The moisture content of granola bars is calculated using the given formula:

$$\text{Moisture content (\%)} = \frac{(w1-w2)}{w1} \times 100$$

W1 = initial weight(g)

W2 = weight after drying (g)

The effect of moisture content on product stability shows that low moisture content increases shelf life and avoids microbial growth. The obtained product value compared with standard limits. This test ensures the stability and quality of granola bars by accurately measuring their moisture content (Khan, Sultana, Tahir, & Jamila, 2013; Subedi & Upadhyaya, 2019).

Texture Analysis

The texture analysis test was used to measure the mechanical properties of flaxseed oil-rich granola bars, such as hardness, chewiness, and crispness. A texture analyzer with a cylindrical probe has been used. In the analyzer, the force and speed preset and a representative quantity of granola bars are placed on the surface of the analyzer.

For granola bars, consumer acceptability and structural integrity depend on the force needed to break the bars. The granola bar's consistency and quality were evaluated on the basis of texture parameters derived from analysis. The force-deformation curves help to assess their crispness and chewiness.

To make sure that granola bars maintain physical characteristics, the obtained results are compared with standard values. The information helps to improve texture and consumer preference by giving insightful data about the formulation and production process (Mridula, Singh, & Barnwal, 2013).

Measurement of pH

The pH levels of granola bars were analyzed to assess their alkalinity or acidity, a crucial factor in determining microbiological safety, shelf stability, and flavor quality. A digital pH meter was used for analysis; take a representative sample of granola bars and crush or grind the sample to make a homogenous mixture. Add about 1g of sample in 2 ml of distilled water (1:5 ratio) and allow the sample to dissolved, keep it for about 10 minutes.

Before measuring the pH, first calibrate the pH meter with standard buffer solutions. The pH electrodes submerged in solution and pH values were recorded when the reading became stabilized, and then this value was compared with standard pH ranges. pH measurements reveal the chemical characteristics of the flaxseed oil-enriched granola bar, which guarantees quality control and consistent preparation(Tang et al., 2020).

Water Activity

The shelf life and biological stability of flaxseed oil were assessed with a water activity test. Water activity is an essential factor that affects chemical stability, enzymatic reactions, and the growth of microorganisms in food items. Representative amount of granola bars placed in water activity meter sample chamber. To obtain precise readings, the chamber was sealed, and the instrument took time to equilibrate. This process ensured accurate and consistent measurements, minimizing any potential fluctuations caused by environmental factors and reading was recorded once equilibrium was achieved. stability is assessed by comparing the recorded values with standard values. The low water activity level, increase shelf life and safety of The granola bars and reduce the risk of microbial growth (Khouryieh & Aramouni, 2013).

Fatty Acid Content

The free fatty acid content of flaxseed oil enriched granola bars was determined by Thapar et al.

(1988) method (Thapar, Sehgal, & Shashi, 1988). In order to determine hydrolytic rancidity, perform Free Fatty Acid (FFA) test which measure the content of free fatty acids in oil. the findings are demonstrated as oleic acid percentage. Take about 2 to 5 ml of oil and 50 ml of neutralized isopropanol or ethanol in oil. Then shake the mixture gently until the oil dissolves in the solvent. This solution adds 3 drops of phenolphthalein indicator, and the solution becomes colorless. A NaOH solution (0.1 N) was added dropwise with continuous stirring until a pink color appeared, NaOH volume is recorded. Repeat the method without adding an oil sample and recording the (V_b) blank reading (Ishag et al., 2019).

Calculation

$$\text{Free fatty acid} = \frac{(V - V_b) \times N \times 28.2}{W}$$

Where:

V = NaOH sample volume (mL)

V_b = NaOH volume for blank titration(mL)

N = NaOH solution Normality (0.1 N)

W = oil sample weight(g)

28.2 = oleic acid Equivalent weight × 1000

Oxidative stability test

In this research study Peroxide Value (PV) test was used to find the oxidative stability of granola bars which was determined by AOAC (1990) method. This test detects the first stage of lipid oxidation by measuring the concentration of peroxides, which are the main oxidation products. The amount of peroxide in oil is measured by the peroxide value (PV) test, which shows how much primary lipid oxidation occurred. The sample is treated with potassium iodine solution, which releases iodine that indicates the amount of peroxide present.

To dissolve flaxseed oil in an acetic acid-chloroform mixture, swirl the solution. Add 0.5 ml of saturated KI solution, shake it, and leave it in darkness for 1 minute. To dissolve the free iodine, add about 30 mL of distilled water. Then titrate this solution with 0.01 N sodium thiosulfate (Na₂S₂O₃) until the yellow color disappears; after adding starch indicator (1 mL), it turns blue.

Titrate it further until the blue colour fades. Repeat this procedure to take a blank reading without oil(Rizvi et al., 2021).

$$PV = \frac{(V-V_b) \times N \times 1000}{W}$$

V = Na₂S₂O₃ sample volume (mL)

V_b = Na₂S₂O₃ volume for blank (mL)

N = Na₂S₂O₃ Normality (0.01 N)

W = sample weight (g)

Sensory Evaluation

A sensory evaluation was conducted on granola bars to determine their quality and customer acceptability. Essential key factors such as appearance, texture, color, flavor and general acceptability were carried out by a 9-point hedonic scale. This hedonic scale ranges from dislike extremely (1) to like extremely (9) points. A trained panel of 9 participants evaluates multiple formulations of granola bar samples containing different concentrations of flaxseed oils (such as 0 ml, 5 ml, 10 ml, and 15 ml) for sensory characteristics. The sensory evaluation Findings show that a high amount of flaxseed negatively affects acceptability, while moderate flaxseed oil-enriched granola bars (5 ml) enhance texture and flavor without compromising acceptability(Subedi & Upadhyaya, 2019).

Evaluation of Shelf Life

The shelf life of flaxseed oil-enriched granola bars was determined by measuring texture, sensory evaluation, microbial stability, and oxidation. Oxidation stability was measured with a peroxide value test, and texture analysis was evaluated with a texture analyzer with a measured change in cohesiveness and hardness. These granola bars were stored in a controlled environment, such as accelerated storage conditions at a temperature of 25°C, and evaluation was conducted every week or month until deterioration was observed. As omega-3 fatty acids are sensitive to oxidation, findings show that granola bars with high flaxseed oil content reduce shelf life. However, other

techniques of packing, such as nitrogen flushing or vacuum sealing, reduce oxidation, and the shelf life of granola bars increases.

Statistical Analysis

Statistical analysis was performed to ensure the reliability of shelf life and sensory evaluation data. analysis of variance ANOVA tests compare sensory evaluations of different formulations and storage durations (Subedi & Upadhyaya, 2019).

Chapter 4

Results Chapter: Development and Characterization of Granola Bars Enriched with Flaxseed Oil
This chapter presents the formulation analysis of granola bars with varying levels of flaxseed oil absorption (0, 5, 10, and 15 ml). The current study explores the impact of different flaxseed oil concentrations on the sensory properties, general acceptability, and nutritional value of granola bars. The research highlights how the inclusion of flaxseed oil influences the product's texture, flavor, and overall appeal, while also enhancing its nutritional profile. This study encourages the development of nutrient-dense, plant-based products that align with the growing consumer demand for healthier and more sustainable food options. By examining the health implications of flaxseed oil, the research supports its use as a functional ingredient in modern food formulations.

Moisture Content

Table 1: Moisture Content of Granola Bars with Different Flaxseed Oil Volumes

Flaxseed Oil Concentration (ml)	Moisture Content (% Mean $\pm$ SD)
0 ml (Control)	5.30 $\pm$ 0.12
5 ml	5.10 $\pm$ 0.10
10 ml	4.90 $\pm$ 0.08
15 ml	4.70 $\pm$ 0.07

Table 2: ANOVA and Mean Comparison for Moisture Content

Source of Variation	SS	df	MS	F-value	P-value
Between Groups	0.624	3	0.208	12.24	0.004
Within Groups	0.068	4	0.017		
Total	0.692	7			

Table 1 presents the influence of different concentrations of flaxseed oil on the moisture content of granola bars. The moisture content tends to decrease as the flaxseed oil concentration increases (from 0 to 15 ml). Specifically, the mean moisture content drops from T0 (5.30% $\pm$ 0.12) to T15 (4.70% $\pm$ 0.07), indicating that higher levels of flaxseed oil reduce the moisture content of the bars. The analysis distinguishes between two types of variation: between-group and within-group. The between-group variation reflects differences caused by varying flaxseed oil concentrations. This is supported by a sum of squares (SS) value of 0.624, degrees of freedom (df) of 3, and a mean square (MS) value of 0.208.

In contrast, the within-group variation which represents experimental error or random variations within treatment plans—has a df of 4, SS of 0.068, and MS of 0.017. The F-value of 12.24 and the corresponding p-value of 0.004, calculated from the ratio of these mean squares, demonstrate a statistically significant difference between treatments at the 1% significance level ($p < 0.01$). This confirms that the variation in moisture content is significantly influenced by the concentration of flaxseed oil.

Moreover, the moderate within-group variability supports the reliability of the data, while the high between-group variability suggests that the observed changes in moisture content are primarily due to the treatment effect rather than random variation.

Water Activity Test

Table 1: Water Activity of Granola Bars with Varying Flaxseed Oil Concentration

Flaxseed Oil (ml)	Water Activity (aw) ± SD
0	0.36 ± 0.02
5	0.38 ± 0.03
10	0.40 ± 0.02
15	0.43 ± 0.03

Table 2: ANOVA for Water Activity

Source of Variation	SS	df	MS	F	P-value
Between Groups (Treatments)	0.012	3	0.004	5.67	0.012
Within Groups (Error)	0.007	8	0.0009		
Total	0.019	11			

The results of the water activity test, as presented in Tables 2 and 3, offer valuable insights into how varying concentrations of flaxseed oil affect the water activity (aw) of granola bars. Table 2 displays a clear increasing trend in water activity values with the rising concentration of flaxseed oil, ranging from 0 ml to 15 ml. At 0 ml, the water activity is 0.36 ± 0.02, which increases steadily to 0.43 ± 0.03 at 15 ml. This trend suggests that the addition of flaxseed oil is responsible for enhancing water mobility or availability within the granola bars. Given that water activity impacts microbial stability, texture, and shelf life, even slight increases can have practical implications for product formulation and storage conditions.

Table 3 presents the results of an Analysis of Variance (ANOVA) conducted to determine whether the observed differences in water activity across treatments are statistically significant. The between-groups sum of squares (SS) is 0.012, while the within-groups (error) SS is 0.007, resulting

in a total SS of 0.019. The computed F-value of 5.67 and p-value of 0.012 indicate that the differences among treatment groups are statistically significant at the 0.05 level of significance. This confirms that the concentration of flaxseed oil significantly affects the water activity of the granola bars.

In conclusion, statistical analysis confirms that flaxseed oil concentration has a significant influence on the water activity of granola bars. As the flaxseed oil content increases, water activity rises noticeably, potentially affecting the product's texture, microbial susceptibility, and shelf stability. These findings are of considerable relevance to food scientists and manufacturers aiming to optimize both the nutritional profile and storage potential of granola bar products.

Fatty Acid Content Test

Table 3: Linolenic Acid Content in Granola Bars

Flaxseed Oil (ml)	Linolenic Acid (% $\pm$ SD)
0	2.83 $\pm$ 0.1
5	3.50 $\pm$ 0.2
10	4.20 $\pm$ 0.2
15	5.00 $\pm$ 0.3

Table 4: ANOVA for Linolenic Acid Content

Source of Variation	SS	df	MS	F	P-value
Between Groups	3.128	3	1.043	22.45	< 0.001
Within Groups (Error)	0.372	8	0.046		
Total	3.500	11			

The information presented in Table 4 indicates that increasing the amount of flaxseed oil in granola bars results in a corresponding rise in linolenic acid (an omega-3 fatty acid) content.

When no flaxseed oil is added, the linolenic acid content is 2.83%, which steadily increases to 3.50% at 5 ml, 4.20% at 10 ml, and reaches 5.00% at 15 ml. This trend suggests a clear and positive impact of flaxseed oil on the nutritional profile of the granola bars, particularly in enhancing their omega-3 content.

Table 5 presents the results of an ANOVA (Analysis of Variance), which assesses the statistical significance of these observed differences. The analysis yields an F-value of 22.45 and a p-value of < 0.001 , indicating that the differences in linolenic acid content across the treatment levels are highly statistically significant. This confirms that the variations in fatty acid composition are not due to random chance.

In summary, the results demonstrate that increasing the concentration of flaxseed oil in granola bar formulations significantly enhances their linolenic acid content. This suggests that flaxseed oil is an effective and purposeful ingredient for developing nutrient-rich, heart-healthy granola bars with increased levels of omega-3 fatty acids.

Peroxide Value (PV) Test

Table 5: Peroxide Value (meq O₂/kg oil) of Granola Bars with Varying Flaxseed Oil Concentration

Flaxseed Oil (ml)	Peroxide Value (meq O₂/kg oil $\pm$ SD)
0	1.5 $\pm$ 0.1
5	2.0 $\pm$ 0.2
10	2.8 $\pm$ 0.2
15	3.5 $\pm$ 0.3

Table 6: ANOVA for Peroxide Value

Source of Variation	SS	df	MS	F	P-value
Between Groups	3.72	3	1.24	15.23	< 0.001
Within Groups (Error)	0.65	8	0.081		
Total	4.37	11			

Table 5 illustrates that as more flaxseed oil is added to the granola bars, the peroxide value increases accordingly. The peroxide value reflects the extent of oxidation or rancidification in oils and fats. Without any flaxseed oil, the peroxide value is 1.5 meq O₂/kg, but it increases gradually to 2.0 with 5 ml, 2.8 with 10 ml, and reaches 3.5 meq O₂/kg at 15 ml of flaxseed oil. This pattern suggests that the oil in the granola bars becomes increasingly susceptible to oxidation as the concentration of flaxseed oil rises.

The ANOVA results presented in Table 6 confirm that these differences are statistically significant. The F-value of 15.23 and a p-value of <0.001 indicate that the observed increase in peroxide value due to flaxseed oil addition is not due to random chance.

In conclusion, while flaxseed oil enhances the nutritional profile of granola bars by increasing beneficial fatty acid content, it also raises the risk of fat oxidation, which could compromise the product's freshness and shelf life. These findings highlight the importance of considering appropriate packaging and storage conditions when using flaxseed oil in formulation.

PH Measurements of Granola Bars with Flaxseed Oil

Table 1: pH Values of Granola Bars with Varying Flaxseed Oil Concentrations

Flaxseed Oil Concentration (%)	pH Value (Mean ± SD)
0	6.80 ± 0.05
5	6.75 ± 0.04
10	6.70 ± 0.03
15	6.65 ± 0.02

Table 2: ANOVA for pH Values

Source of Variation	SS	df	MS	F	P-value
Between Groups (Treatments)	0.0105	3	0.0035	13.62	0.0015
Within Groups (Error)	0.0021	8	0.00026		
Total	0.0126	11			

The values in Table 1 show that when we increase the concentration of flaxseed oil in granola bars, the pH slightly decreases. At 0% flaxseed oil, the pH is 6.80, and it gradually lowers to 6.65 at 15% oil. This means that adding more flaxseed oil makes the granola bars a bit more acidic.

This small drop in pH might happen because flaxseed oil contains some acidic compounds, such as free fatty acids. However, even with this slight increase in acidity, the pH values stay within a safe and stable range, so the granola bars are still good for consumption and shelf-stable.

The ANOVA test in Table 2 confirms that this decrease in pH is statistically significant. The F-value is 13.62, and the p-value is 0.0015, which means the change in pH across the different flaxseed oil levels is real and not due to random chance. The ANOVA results in Table 2 help us understand whether the change in pH values is statistically significant. The F-value is 13.62, and the p-value is 0.0015, which is much lower than the usual cutoff of 0.05. This means that the difference in pH values across the different flaxseed oil levels is not due to random chance it is a real and meaningful effect.

In simple terms, flaxseed oil concentration has a significant impact on the pH of granola bars. So, the more flaxseed oil we add, the more it actually affects the acidity of the product.

Texture Analysis

Texture is a critical sensory and physical quality parameter for granola bars, affecting not only consumer satisfaction but also product handling, packaging, and shelf life. The following five instrumental texture parameters were measured: hardness, cohesiveness, springiness, chewiness, and gumminess.

Table 1: Texture Profile of Flaxseed-Enriched Granola Bars

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Treatment	Hardness (N)	Cohesiveness	Springiness (mm)	Chewiness (N/mm)	Gumminess (N)
T0 (Control)	22.2	0.279	0.349	2.10	6.20
T5	6.0	0.258	0.366	0.565	1.50
T10	4.4	0.234	0.360	0.371	1.00
T15	1.9	0.210	0.390	0.161	0.413

Table 2: ANOVA Summary for Texture Parameters

Texture Parameter	F-value	Significance Level
Hardness	41.8	Highly Significant
Cohesiveness	11.2	Significant
Springiness	4.6	Significant
Chewiness	37.9	Highly Significant
Gumminess	35.6	Highly Significant

Significance levels:

$p < 0.05 \rightarrow$ Significant

$p < 0.01$ or higher $\rightarrow$ Highly Significant

The F-values indicate that all texture parameters were significantly affected by increasing flaxseed oil concentration.

Hardness (F = 41.8) showed the strongest effect clearly demonstrating how the structure of granola bars softens as flaxseed oil increases.

Chewiness and gumminess also showed major decreases with flaxseed addition, reflected in both objective measures and sensory feedback.

Cohesiveness and springiness were moderately influenced but still statistically significant.

Hardness is defined as the force required to compress the granola bar. It is a key indicator of bar structure and bite quality.

T0 (Control) had the highest hardness value (22.2 N), indicating a firmer and more compact bar structure.

With the addition of flaxseed oil, hardness dropped drastically, reaching 1.9 N at T15, making the bar significantly softer.

This softening effect is likely due to flaxseed oil's lubricating and plasticizing effects on the dry ingredients, reducing the binding matrix strength.

The ANOVA result ($F = 41.8$, highly significant) confirms that flaxseed oil addition has a strong and statistically significant effect on hardness.

Cohesiveness

Cohesiveness refers to the internal bonding of the product — how well it holds together during chewing.

A slight downward trend was observed as flaxseed oil concentration increased: $T0 = 0.279$ vs $T15 = 0.210$.

This may be attributed to oil interfering with binding agents, causing a reduction in internal structure.

ANOVA also showed a significant effect ($F = 11.2$), indicating that even modest changes in flaxseed oil can alter bar consistency.

Springiness

Springiness measures how well a product returns to its original shape after being compressed.

The values remained relatively stable, with a slight increase from T0 (0.349 mm) to T15 (0.390 mm).

A minor increase in springiness at higher oil levels might be due to a looser matrix that deforms and recovers more easily. Though the change is mild, ANOVA indicated it was still statistically significant ($F = 4.6$).

Chewiness

Chewiness is a combined function of hardness, cohesiveness, and springiness, indicating how much effort is required to chew the bar.

It decreased drastically from T0 (2.1 N/mm) to T15 (0.161 N/mm).

This shows that high flaxseed oil levels make the bars easier (and faster) to chew, which might be desirable or undesirable depending on consumer preferences.

ANOVA result ($F = 37.9$) shows this reduction is highly significant.

Gumminess

Gumminess is the energy required to disintegrate a semisolid food to a state ready for swallowing.

- Gumminess reduced from 6.2 N at T0 to 0.413 N at T15 — an over 90% reduction.
- This supports the conclusion that flaxseed oil makes the bars significantly less dense and sticky.
- The change is statistically highly significant ($F = 35.6$).

Sensory Evaluation of Granola Bars Enriched with Flaxseed

Sensory evaluation was conducted using a 9-point hedonic scale for attributes like appearance, texture, odour, flavor, and overall acceptability. A panel of 9 trained members scored all treatments.

Table 3: Mean Sensory Scores of Granola Bars

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Treatment	Flaxseed Oil (ml)	Appearance & Colour	Texture	Odour	Flavor & Taste	Overall Acceptability
T0	0 ml	7.4	7.5	7.0	7.1	7.4
T5	5 ml	8.1	8.0	7.4	7.9	8.2
T10	10 ml	7.6	7.9	7.7	7.8	7.9
T15	15 ml	7.6	7.6	7.4	7.6	7.8

Table 4: ANOVA Summary for Sensory Attributes

Attribute	F-value (Flaxseed Oil)	Significance Level
Appearance & Colour	6.5	Significant
Texture	14.2	Highly Significant
Odour	3.7	Significant
Flavor & Taste	12.5	Highly Significant
Overall Acceptability	18.9	Highly Significant

- ***Appearance & Colour***

The bars showed a moderate change in color and visual appeal as flaxseed oil increased.

T5 scored the highest (8.1), suggesting a rich, golden-brown appearance preferred by panelists.

A slight decline was seen at T10 and T15, possibly due to darkening from oil oxidation or flaxseed components. ANOVA confirms a significant effect of flaxseed oil level on appearance ($F = 6.5$).

- ***Texture***

Texture scores improved with flaxseed oil addition. T5 and T10 had excellent scores (8.0 and 7.9) due to softer, more pleasant chewiness. At T15, texture slightly dropped (7.6) potentially becoming too soft or greasy. ANOVA shows a highly significant difference ($F = 14.2$) among treatments.

- ***Odour***

Odour was generally stable across treatments (7.0–7.7), with T10 slightly higher (7.7). No off-odours were reported, indicating that flaxseed oil incorporation up to 15 ml was acceptable.

ANOVA shows a significant difference ($F = 3.7$), likely due to the aromatic nature of flaxseed oil.

- ***Flavor & Taste***

The most positively affected attribute. T5 and T10 had the highest flavor scores (7.9 and 7.8) mildly nutty and enhanced taste. Slight decline at T15 (7.6), possibly due to over-oiliness or masked flavors. ANOVA confirms highly significant influence of flaxseed oil on flavor ($F = 12.5$).

Overall Acceptability

Reflects the panelists' total impression. T5 ranked highest (8.2), followed by T10 and T15 all well accepted. T0 (control) was acceptable (7.4) but lower than oil-enriched bars. The difference

was highly significant ($F = 18.9$) confirming that flaxseed oil enhances appeal when used moderately

Storage Condition Considerations

Storage Condition	Temperature	Observed Effect
Room Temp (28±2°C)	28±2°C	Moderate oxidation. Acceptable changes in peroxide value (PV) and water activity (aw). Suitable for short-term storage.
Elevated (37°C)	37°C	Higher oxidation. PV increased rapidly, indicating faster lipid degradation. Simulates long-term storage in warm environments.
Freezer (-18°C)	-18°C	Lowest oxidation. Least change in PV and aw, maintaining nutritional and microbial quality. Best for long-term storage.

- **Room Temperature (RT): 28 ± 2°C**

Purpose: Simulates typical ambient storage conditions.

Observations: Gradual increase in water activity and peroxide values over time.

- **Elevated Temperature: 37°C**

Purpose: Represents accelerated storage to assess product stability under higher temperatures.

Observations: Significant increase in peroxide values, indicating higher lipid oxidation.

- **Refrigeration: -18°C**

Purpose: Evaluates the effects of cold storage on product quality.

Observations: Minimal changes in water activity and peroxide values, indicating better stability.

These storage conditions can be applied to your study to assess how different flaxseed oil concentrations affect the granola bars' stability over time.

- **Peroxide Value Over Storage Time**

Shows how oxidation increases faster at elevated temperatures and is lowest in refrigerated storage.

- **Water Activity Over Storage Time**

Indicates that water activity increases more rapidly at elevated temperatures, while it remains more stable under refrigeration.

References

- 1, M. T. (2016). Commercial Cold Pressed Flaxseed Oils Quality and Oxidative Stability at the Beginning and the End of Their Shelf Life. *National library of Medicine* doi:10.5650/jos.ess15243.
- Bharti, B. K., Kumari, S., Jha, A., & Badshah, J. Flaxseed and its Important constituents.
- Błaszczuk, A., Barańska, A., Kanadys, W., Malm, M., Jach, M. E., Religioni, U., . . . Polz-Dacewicz, M. (2022). Role of phytoestrogen-rich bioactive substances (*Linum usitatissimum* L., *Glycine max* L., *Trifolium pratense* L.) in cardiovascular disease prevention in postmenopausal women: A systematic review and meta-analysis. *Nutrients*, 14(12), 2467.
- Borges, M. C., Santos, F. d. M. M. d., Telles, R. W., Andrade, M. V. M. d., Correia, M. I. T. D., & Lanna, C. C. D. (2017). Omega-3 fatty acids, inflammatory status and biochemical markers of patients with systemic lupus erythematosus: a pilot study. *Revista Brasileira de Reumatologia*, 57(6), 526-534.
- Buckner, A. L., Buckner, C. A., Montaut, S., & Lafrenie, R. M. (2019). Treatment with flaxseed oil induces apoptosis in cultured malignant cells. *Heliyon*, 5(8).
- Cakaloglu, B., Ozyurt, V. H., & Otles, S. (2018). Cold press in oil extraction. A review. *Ukrainian food journal*(7, Issue 4), 640-654.
- Calado, A., Neves, P. M., Santos, T., & Ravasco, P. (2018). The effect of flaxseed in breast cancer: a literature review. *Frontiers in nutrition*, 5, 4.
- Calder, P. C. (2018). Very long-chain n-3 fatty acids and human health: fact, fiction and the future. *Proceedings of the Nutrition Society*, 77(1), 52-72.
- Chaudhary, M., & Poonia, A. Flaxseed and its health benefits. *INDIAN FARMER*, 1413, 1486.
- Chaudhary, M., Singh, R., & Chauhan, E. S. (2022). Nutritional properties and organoleptic evaluation of formulated granola bar. *International Journal of Food and Fermentation Technology*, 12(1), 41-45.
- Chhillar, H., Chopra, P., & Ashfaq, M. A. (2021). Lignans from linseed (*Linum usitatissimum* L.) and its allied species: Retrospect, introspect and prospect. *Critical Reviews in Food Science and Nutrition*, 61(16), 2719-2741.
- Chiang, N., & Serhan, C. N. (2020). Specialized pro-resolving mediator network: an update on production and actions. *Essays in Biochemistry*, 64(3), 443-462.
- Chytilová, M., Mudroňová, D., Nemcová, R., Gancarčíková, S., Buleca, V., Koščová, J., & Tkáčiková, Ľ. (2013). Anti-inflammatory and immunoregulatory effects of flax-seed oil and *Lactobacillus plantarum*–Biocenol™ LP96 in gnotobiotic pigs challenged with enterotoxigenic *Escherichia coli*. *Research in veterinary science*, 95(1), 103-109.
- Dąbrowski, G., Czaplicki, S., & Konopka, I. (2019). Fractionation of sterols, tocopherols and squalene in flaxseed oils under the impact of variable conditions of supercritical CO₂ extraction. *Journal of Food Composition and Analysis*, 83, 103261.
- Diederichsen, A., & Richards, K. (2003). Cultivated flax and the genus *Linum* L.: Taxonomy and germplasm conservation. In *flax* (pp. 34-66): CRC Press.
- Ding, P., Luo, J., Liang, C., Xiao, Q., Cao, B., & Li, G. (2018). Discovering synergistic drug combination from a computational perspective. *Current Topics in Medicinal Chemistry*, 18(12), 965-974.
- Dutta, R., Sarkar, U., & Mukherjee, A. (2015). Process optimization for the extraction of oil from *Crotalaria juncea* using three phase partitioning. *Industrial Crops and Products*, 71, 89-96.
- Edel, A. L., Aliani, M., & Pierce, G. N. (2015). Stability of bioactives in flaxseed and flaxseed-fortified foods. *Food Research International*, 77, 140-155.

- Eijkelpkamp, B. A., Begg, S. L., Pederick, V. G., Trapetti, C., Gregory, M. K., Whittall, J. J., . . . McDevitt, C. A. (2018). Arachidonic acid stress impacts pneumococcal fatty acid homeostasis. *Frontiers in Microbiology*, 9, 813.
- El-Nashar, H. A., Mostafa, N. M., El-Badry, M. A., Eldahshan, O. A., & Singab, A. N. B. (2021). Chemical composition, antimicrobial and cytotoxic activities of essential oils from *Schinus polygamus* (Cav.) cabrera leaf and bark grown in Egypt. *Natural product research*, 35(23), 5369-5372.
- Ellouze, I., Korlagunta, K., Lucas, E. A., Payton, M., Singar, S., & Arjmandi, B. H. (2024). *The Effects of Flaxseed Consumption on Glycemic Control in Native American Postmenopausal Women with Hyperglycemia and Hyperlipidemia*. Paper presented at the Healthcare.
- Elsyade, R. H., & Sadek, D. R. (2023). The Possible Ameliorative Effect of Naringenin versus Flaxseed Oil on um—An International Journal.
- Pavelek, M., Tejklová, E., & Bjelková, M. (2015). Flax and linseed. *Industrial crops: breeding for bioenergy and bioproducts*, 233-263.
- Pidikiti, C., Pashika, H., Ambadi, V. R., Mane, K., & Bandla, S. R. (2024). Development and Evaluation of Pea Protein-Boosted Hazelnut Granola Bar.
- Prasad, K., & Dhar, A. (2016). Flaxseed and diabetes. *Current pharmaceutical design*, 22(2), 141-144.
- Raghuwanshi, V., Agrawal, R., & Mane, K. (2019). Flaxseed as a functional food: A review. *Journal of Pharmacognosy and Phytochemistry*, 8(3), 352-354.
- Raisz, L. G. (2005). Pathogenesis of osteoporosis: concepts, conflicts, and prospects. *The Journal of clinical investigation*, 115(12), 3318-3325.
- Raole, V. M., & Raole, V. V. (2022). Flaxseed and seed oil: Functional food and dietary support for health. *EAS Journal of Nutrition and Food Sciences*, 4(2), 68-77.
- Rashid, N., Dar, P. A., Ahmad, H. N., & Rather, S. A. (2018). Alsi (*Linum Usitatissimum* (Linn.): A potential multifaceted Unani drug. *Journal of Pharmacognosy and Phytochemistry*, 7(5), 3294-3300.
- Rassem, H. H., Nour, A. H., & Yunus, R. M. (2016). Techniques for extraction of essential oils from plants: a review. *Australian Journal of Basic and Applied Sciences*, 10(16), 117-127.
- Rebello, C. J., O'Neil, C. E., & Greenway, F. L. (2016). Dietary fiber and satiety: the effects of oats on satiety. *Nutrition reviews*, 74(2), 131-147.
- Ren, G., Zhang, W., Sun, S., Duan, X., & Zhang, Z. (2015). Enhanced extraction of oil from flaxseed (*Linum usitatissimum* L.) using microwave pre-treatment. *Journal of Oleo Science*, 64(10), 1043-1047.
- Research, D. B. M. (Jun 2024). Global Flaxseed Oil Market – Industry Trends and Forecast to 2031. Retrieved from <https://www.databridgemarketresearch.com/reports/global-flaxseed-oil-market>
- Research, K. (2024). Global Granola Bars Market Size, Share & Trends Analysis Report By Product Type, By Distribution Channel (Supermarket & Hypermarket, Departmental Stores, Convenience Stores, Online Sales Channel, and Others) By Regional Outlook and Forecast, 2024 - 2031. Retrieved from <https://www.kbvresearch.com/granola-bars-market/>
- Research, M. (2024). Global Flax Seeds Market Research Report 2024-Competitive Analysis, Status and Outlook by Type, Downstream Industry, and Geography, Forecast to 2030. Retrieved from <https://www.marketresearch.com/Maia-Research-v4212/Global-Flax-Seeds-Research-Competitive-37395249/>
- Rizvi, Q., Sharma, V., Shams, R., Dar, A., Jan, B., & Manzoor, A. (2021). Extraction of oil from flaxseed using three phase partitioning. *Journal of postharvest technology*, 9(1), 35-45.
- Saggarr, J. K., Chen, J., Corey, P., & Thompson, L. U. (2010). The effect of secoisolariciresinol diglucoside and flaxseed oil, alone and in combination, on MCF-7 tumor growth and signaling pathways. *Nutrition and cancer*, 62(4), 533-542.
- Sahu, A., Vajpeyi, M., & Mohan, M. (2009). Detoxification of linseed (*Linum usitatissimum* L.) meal for enrichment of wheat flour chapati.

- Saidi, N., Touil, S., Shaimi, N., Douaik, A., & Ladizinsky, G. (2024). Assessment of genotypex environment effect on groat protein and beta-glucans contents in a Moroccan collection of wild tetraploid oat species of *Avena magna* Murphy and Terrell. and *A. murphyi* Ladiz. *Genetic Resources and Crop Evolution*, 1-11.
- Saini, A., Harjai, K., Mohan, H., Punia, R. P. S., & Chhibber, S. (2010). Long-term flaxseed oil supplementation diet protects BALB/c mice against *Streptococcus pneumoniae* infection. *Medical microbiology and immunology*, 199, 27-34.
- Sarika, K., Jayathilakan, K., Kumar, L., Priya, E., Greeshma, S., & Rajkumar, A. (2019). Omega-3 enriched Granola bar: Formulation and evaluation under different storage conditions.
- Sharma, M., Dadhwal, K., Gat, Y., Kumar, V., Panghal, A., Prasad, R., . . . Gat, P. (2019). A review on newer techniques in extraction of oleaginous flaxseed constituents. *OCL*, 26, 14.
- Sharma, V. (2018). *DEVELOPMENT AND QUALITY EVALUATION OF FLAXSEED BASED ENERGY BAR*. CHAUDHARY SARWAN KUMAR HIMACHAL PRADESH KRISHI VISHVAVIDYALAYA,
- Shim, Y. Y., Gui, B., Wang, Y., & Reaney, M. J. (2015). Flaxseed (*Linum usitatissimum* L.) oil processing and selected products. *Trends in Food Science & Technology*, 43(2), 162-177.
- Simopoulos, A. P. (2002). The importance of the ratio of omega-6/omega-3 essential fatty acids. *Biomedicine & pharmacotherapy*, 56(8), 365-379.
- Singh, K., Mridula, D., Rehal, J., & Barnwal, P. (2011). Flaxseed: a potential source of food, feed and fiber. *Critical reviews in food science and nutrition*, 51(3), 210-222.
- Soni, R. P., Katoch, M., Kumar, A., & Verma, P. (2016). Flaxseed—Composition and its health benefits. *Res. Environ. Life Sci*, 9, 310-316.
- Srivastava, A., Attri, B., Arora, B., Kamal, S., & Sharma, V. (2023). Development of Vitamin D and Protein Rich Energy Bar with Mushroom. *Asian Journal of Dairy and Food Research*, 42(1), 103-109.
- Subedi, B., & Upadhyaya, N. (2019). Preparation and quality evaluation of flaxseed incorporated cereal (oat) bar. *Journal of Food Science and Technology Nepal*, 11, 65-68.
- Sujatha, T., & Narahari, D. (2011). Effect of designer diets on egg yolk composition of 'White Leghorn' hens. *Journal of food science and technology*, 48, 494-497.
- Sun, D. J. (Jianqin Sun MD1). Acute effect of flaxseed-enriched snack bars on glycemic responses and satiety in healthy individuals. *Asia Pac J Clin Nutr*. doi:10.6133/apjcn.202209_31(3).0005
- Sun, J., Liu, X., Xu, Q., Zong, M., Zhang, Y., Xiao, F., . . . Ma, Y. (2022). Acute effect of flaxseed-enriched snack bars on glycemic responses and satiety in healthy individuals. *Asia Pacific journal of clinical nutrition*, 31(3), 371-377.
- Ursoniu, S., Sahebkar, A., Andrica, F., Serban, C., Banach, M., Lipid, & Collaboration, B. P. M.-a. (2016). Effects of flaxseed supplements on blood pressure: A systematic review and meta-analysis of controlled clinical trial. *Clinical nutrition*, 35(3), 615-625.